ML1516 - CLOUD COMPUTING FOR MACHINE LEARNING

UNIT I - FUNDAMENTALS OF CLOUD COMPUTING

Introduction to Cloud Computing - Characteristics of Cloud Computing - Cloud deployment models: public, private, hybrid, community – Cloud Interoperability and Standards, Scalability and Fault Tolerance – Elasticity in Cloud – On-demand Provisioning – Pros and Cons of cloud computing.

Introduction to Cloud Computing

- Over the last three decades, businesses that use computing resources have learned to face a vast array of buzzwords like grid computing, utility computing, autonomic computing, on-demand computing and so on.
- A new buzzword named cloud computing is presently in state-of-the-art and it is generating all sorts of confusion about what it actually means.
- In history, the term cloud has been used as a metaphor for the Internet.

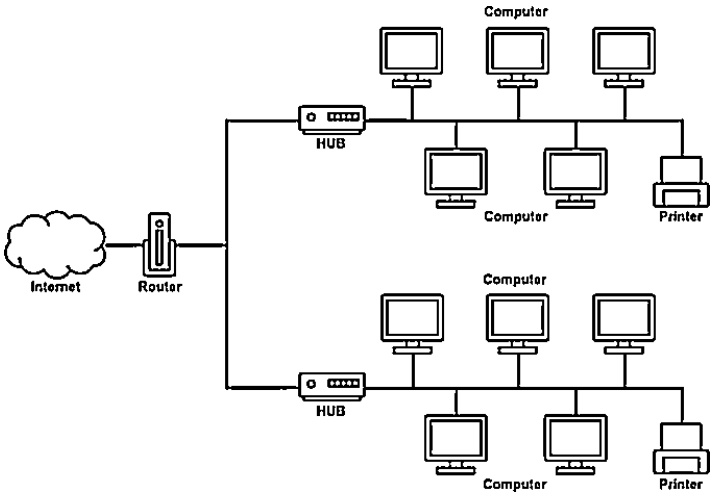


Figure 1.1 illustration of network diagram

- This usage of the term was originally derived from its common illustration in network diagrams as an outline of a cloud and the symbolic representation used to represent the transport of data across the network to an endpoint location on the other side of the network.
- Figure 1.1 illustrates the network diagram which includes the symbolic representation ofcloud
- The cloud computing concepts were initiated in 1961, when Professor John McCarthy suggested that computer time-sharing technology might lead to a future where computing power and specific applications might be sold through a utility-type business model.
- This idea became very popular in the late 1960s, but in mid 1970s the idea vanished away when it became clear that the IT Industries of the day were unable to sustain sucha innovative computing model. However, since the turn of the millennium, the concept has been restored.
- Utility computing is the provision of computational resources and storage resources as a metered service, similar to those provided by a traditional public utility company. This is not a new idea.
- This form of computing is growing in popularity, however, as companieshave begun to extend the

model to a cloud computing paradigm providing virtual serversthat IT departments and users can access on demand.

- In early days, enterprises used the utility computing model primarily for non-mission- critical requirements, but that is quickly changing as trust and reliability issues are resolved.
- Research analysts and technology vendors are inclined to define cloud computing very closely, as a new type of utility computing that basically uses virtual servers that have been made available to third parties via the Internet.
- Others aimed to describe the term cloud computing using a very broad, all-inclusive application of the virtual computing platform. They confront that anything beyond the network firewall limit is in the cloud.
- A more softened view of cloud computing considers it the delivery of computational resources from a location other than the one from which the end users are computing.
- The cloud sees no borders and thus has made the world a much smaller place. Similarto that the Internet is also global in scope but respects only established communication paths.
- People from everywhere now have access to other people from anywhere else.
- Globalization of computing assets may be the major contribution the cloud has made to date. For this reason, the cloud is the subject of many complex geopolitical issues.
- Cloud computing is viewed as a resource available as a service for virtual data centers. Cloud computing and virtual data centers are different one.
- For example, Amazon's S3 is Simple Storage Service. This is a data storage service designed for use across the Internet. It is designed to create web scalable computing easier for developers.
- Another example is Google Apps. This provides online access via a web browser to the most common office and business applications used today. The Google server stores all the software and user data.
- Managed service providers (MSPs) offers one of the oldest form of cloud computing.
- A managed service is an application that is accessible to an organization's IT infrastructure rather than to end users which include virus scanning for email, anti-spam services such as Postini, desktop management services offered by CenterBeam or Everdream, and application performance monitoring.
- Grid computing is often confused with cloud computing. Grid computing is a form of distributed computing model that implements a virtual supercomputer made up of a cluster of networked or Inter networked computers involved to perform very large tasks.
- Most of the cloud computing deployments in market today are powered by grid computing implementations and are billed like utilities, but cloud computing paradigm is evolved next step away from the grid utility model.
- The majority of cloud computing infrastructure consists of time tested and highly reliableservices

built on servers with varying levels of virtualized technologies, which aredelivered via large scale data centers operating under various service level agreements that require 99.9999% uptime.

Definition of cloud

- Cloud computing is a model for delivering IT services in which resources are retrieved from the internet through web based tools and applications rather than a direct connection to the server.
- The cloud is a large group of interconnected computers. These computers can be personal computers or network servers; they can be public or private.
- Cloud computing is a technology that uses the internet and central remote servers to maintain data and applications.
- Eg: Yahoo email or Gmail etc

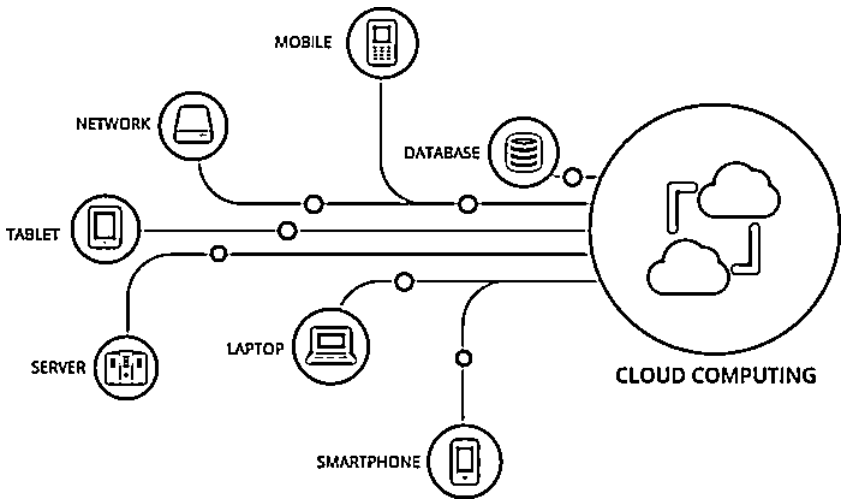


Figure 1.2 Cloud Computing Paradigm

- In other words, cloud computing is a distributed computing model over a network and means the ability to run a program on many connected components at a same time
- In the cloud computing environment, real server machines are replaced by virtual machines. Such virtual machines do not physically exist and can therefore be moved around and scaled up or down on the fly without affecting the cloud user as like a natural cloud.
- Cloud refers to software, platform, and Infrastructure that are sold as a service. The services accessed remotely through the Internet
- The cloud users can simply log on to the network without installing anything. They do not pay for hardware and maintenance. But the service providers pay for physical equipmentand maintenance.
- The concept of cloud computing becomes much more understandable when one begins to think

about what modern IT environments always require scalable capacity or additional capabilities to their infrastructure dynamically, without investing money in the purchase of new infrastructure, all the while without needing to conduct training for newpersonnel and without the need for licensing new software.

- The cloud model is composed of three components.

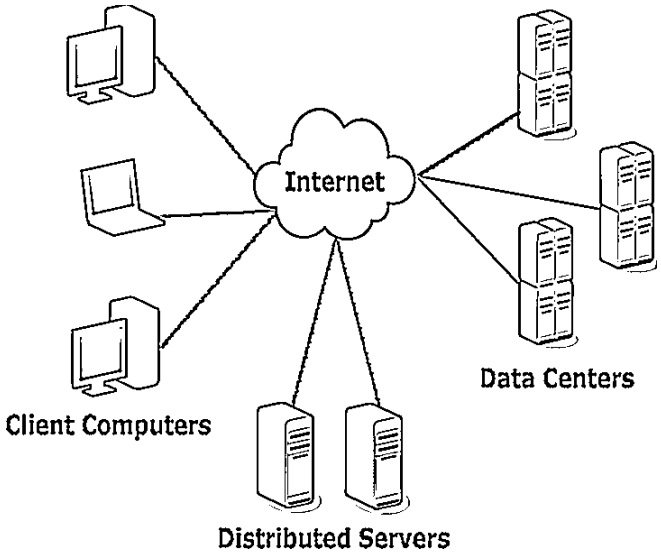


Figure 1.3 Cloud Components

- **Clients** are simple computers might be laptop, tablet, mobile phone.
- Categories of clients are Mobile clients, Thin clients and Thick clients.
- Mobile clients which includes smartphones and PDAs
- Thin clients which include servers without internal hardware. Usage of this typeof clients leads to Low hardware cost, Low IT Cost, Less power consumption andless noise.
- Thick clients which includes regular computers.
- **Data Center** is a collection of servers and it contains clients requested applications.
- **Distributed Server** in which server is distributed in different geographical locations

Evolution of Cloud Computing

- It is important to understand the evolution of computing in order to get an appreciation ofhow IT based environments got into the cloud environment. Looking at the evolution of the computing hardware itself, from the first generation to the fourth generation of computers, shows how the IT industry's got from there to here.
- The hardware is a part of the evolutionary process. As hardware evolved, so did the software. As networking evolved, so did the rules for how computers communicate. The development of such rules or protocols, helped to drive the evolution of Internet software.

- Establishing a common protocol for the Internet led directly to rapid growth in the number of users online.
- Today, enterprises discuss about the uses of IPv6 (Internet Protocol version 6) to ease addressing concerns and for improving the methods used to communicate over the Internet.
- Usage of web browsers led to a stable migration away from the traditional data center model to a cloud computing based model. And also, impact of technologies such as server virtualization, parallel processing, vector processing, symmetric multiprocessing,and massively parallel processing fueled radical change in IT era.

Characteristics of Cloud Computing:

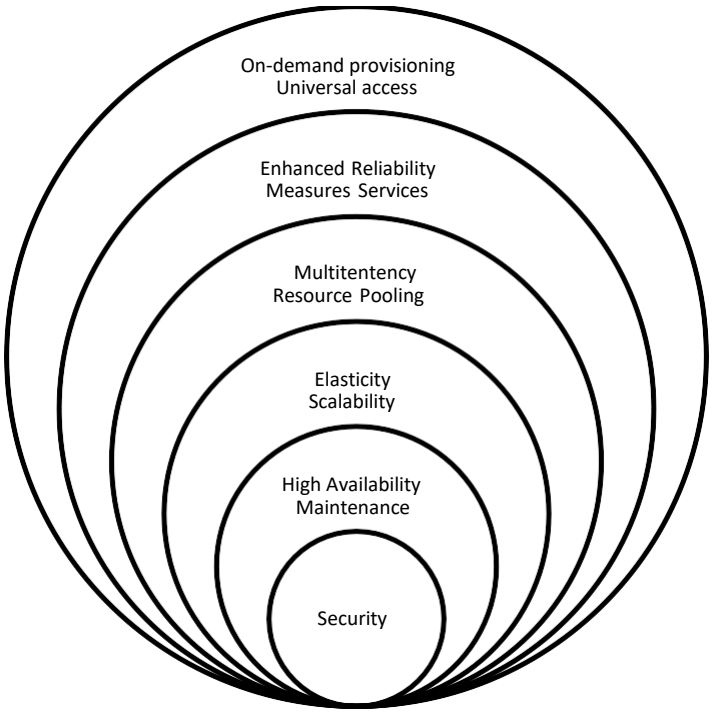
From the cloud computing's various definitions; a certain set of key characteristics emerges. Figure illustrates various key characteristics related to cloud computing paradigm.

On-demand Provisioning

- On-demand provisioning is the single most important characteristic of cloud computing, itallows the users to request or release resources whenever they want.
- These demands are thereafter automatically granted by a cloud provider's service and the users are only charged for their usage, i.e., the time they were in possession of the resources.
- The reactivity of a cloud solution, with regard to resource provisioning is indeed of prime importance as it is closely related to the cloud's pay-as-you-go business model.
- It is one of the important and valuable features of Cloud Computing as the user can continuously monitor the server uptime, capabilities, and allotted network storage. With this feature, the user can also monitor the computing capabilities.

Universal Access

- Resources in the cloud need not only be provisioned rapidly but also accessed and managed universally, using standard Internet protocols, typically via RESTful web services.
- This enables the users to access their cloud resources using any type of devices, provided they have an Internet connection.
- Universal access is a key feature behind the cloud's widespread adoption, not only by professional actors but also by the general public that is nowadays familiar with cloud based solutions such as cloud storage or media streaming.
- Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms such as mobile phones,tablets, laptops, and workstations.



Cloud Characteristics

Enhanced Reliability

- Cloud computing enables the users to enhance the reliability of their applications.
- Reliability is already built in many cloud solutions via storage redundancy.
- Cloud providers usually have more than one data center and further reliability can be achieved by backing data up in different locations.
- This can also be used to ensure service availability, in the case of routine maintenance operations or the rarer case of a natural disaster.
- The user can achieve further reliability using the services of different cloud providers.

Measured Services

- Cloud computing refers generally to paid services.
- The customers are entitled to a certain quality of service, guaranteed by the Service Level Agreement that they should be able to supervise.
- Therefore, cloud providers offer monitoring tools, either using a graphical interface or viaan API.
- These tools also help the providers themselves for billing and management purposes.

Multitenancy

- As the grid before, the cloud's resources are shared by different simultaneous users. These users had to reserve in advance a fixed number of physical machines for a fixed amount of time.

- In virtualized data centers, a user's provisioned resources no longer correspond to the physical infrastructure and can be dispatched over multiple physical machines.
- They can also run alongside other users' provisioned resources thus requiring a lesser amount of physical resources. Consequently, important energy savings can be made by shutting down the unused resources or putting them in energy saving mode.

Resource pooling

- The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
- There is a sense of location independence in that the customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter).
- Examples of resources include storage, processing, memory, and network bandwidth.

Rapid elasticity and Scalability

- Elasticity is the ability of a system to include and exclude resources like CPU cores, memory, Virtual Machine and container instances to adapt to the load variation in real time.
- Elasticity is a dynamic property for cloud computing. There are two types of elasticity. Horizontal and Vertical.
- Horizontal elasticity consists in adding or removing instances of computing resources associated with an application.
- Vertical elasticity consists in increasing or decreasing characteristics of computing resources, such as CPU time, cores, memory, and network bandwidth.
- There are other terms such as scalability and efficiency, which are associated with elasticity but their meaning is different from elasticity while they are used interchangeably in some cases.
- Scalability is the ability of the system to sustain increasing workloads by making use of additional resources, it is time independent and it is similar to the provisioning state in elasticity but the time has no effect on the system (static property).
- The following equation that summarizes the elasticity concept in cloud computing.

  Auto scaling = Scalability +Automation
  Elasticity = Auto scaling + Optimization

- It means that the elasticity is built on top of scalability. It can be considered as an automation of the concept of scalability, however, it aims to optimize at best and as quickly as possible the resources at a given time.
- Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand.
- To the consumer, the capabilities available for provisioning often appear to be unlimited and

can be appropriated in any quantity at any time.

Easy Maintenance

- The servers are easily maintained and the downtime is very low and even in some cases, there is no downtime.
- Cloud Computing comes up with an update every time by gradually making it better. Theupdates are more compatible with the devices and perform faster than older ones along with the bugs which are fixed.

High Availability

- The capabilities of the Cloud can be modified as per the use and can be extended a lot. It analyzes the storage usage and allows the user to buy extra Cloud storage if needed for a very small amount.

Security

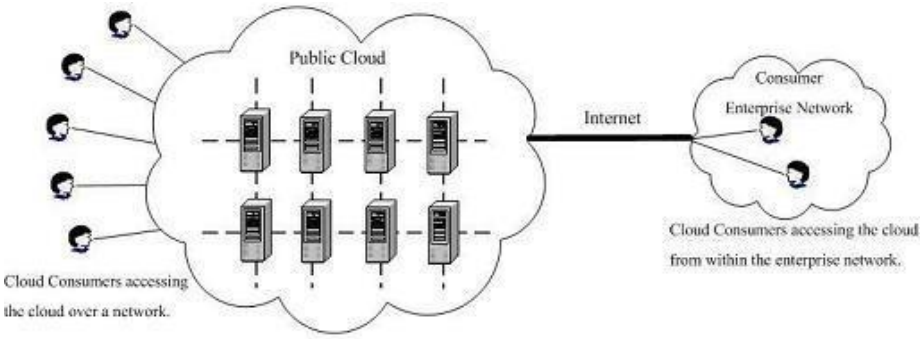
- Cloud Security is one of the best features of cloud computing. It creates a snapshot of the data stored so that the data may not get lost even if one of the servers gets damaged.
- The data is stored within the storage devices, which cannot be hacked and utilized by any other person. The storage service is quick and reliable.

Cloud deployment models:

- As identified in the NIST cloud computing definition, a cloud infrastructure may be operated in one of the following deployment models: public cloud, private cloud, community cloud, or hybrid cloud.
- The differences are based on how exclusive the computing resources are made to a Cloud Consumer.

Public Cloud

- A public cloud is one in which the cloud infrastructure and computing resources are made available to the general public over a public network.
- A public cloud is owned by an organization selling cloud services, and serves a diverse pool of clients



Public Cloud

Benefits of choosing a Public Cloud

- One of the main benefits that come with using public cloud services is near unlimited scalability.
- The resources are pretty much offered based on demand. So any changes in activity level can be handled very easily.
- This in turn brings with it cost effectiveness.
- Public cloud allows pooling of a large number of resources, users are benefiting from thesavings of large scale operations.
- There are many services like Google Drive which are offered for free.
- Finally, the vast network of servers involved in public cloud services means that it can benefit from greater reliability.
- Even if one data center was to fail entirely, the network simply redistributes the load among the remaining enters making it highly unlikely that the public cloud would ever fail.
- In summary, the benefits of the public cloud are:
    - Easy scalability
    - Cost effectiveness
    - Increased reliability

Disadvantages of choosing a Public Cloud

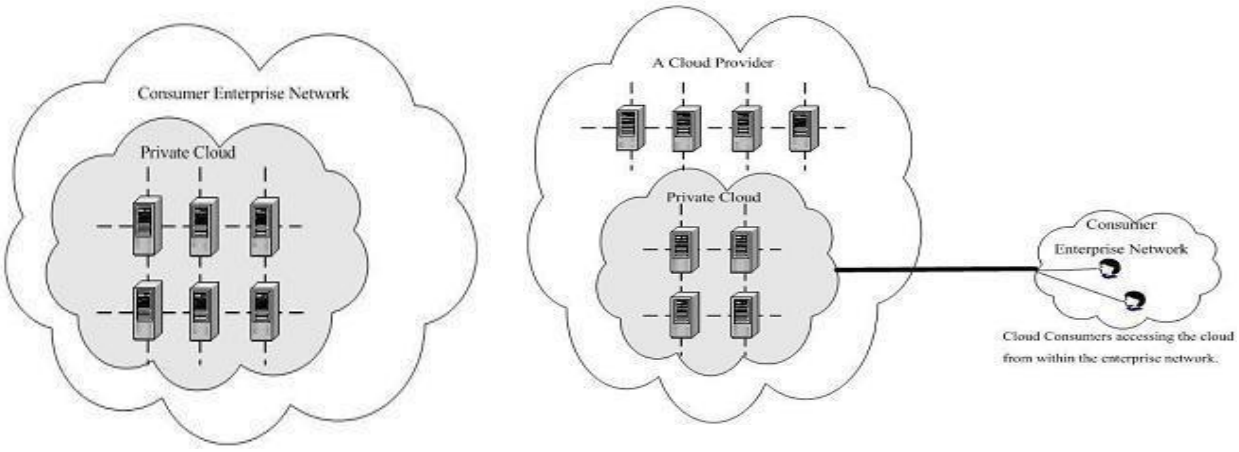
- There are of course downsides to using public cloud services.
- At the top of the list is the fact that the security of data held within a public cloud is a cause for concern.
- It is often seen as an advantage that the public cloud has no geographical restrictions

making access easy from everywhere, but on the flip side this could mean that the server is in a different country which is governed by an entirely different set of security and/or privacy regulations.

- This could mean that your data is not all that secure making it unwise to use public cloud services for sensitive data.

Private Cloud

- A private cloud gives a single Cloud Consumer's organization the exclusive access toand usage of the infrastructure and computational resources.
- It may be managed either by the Cloud Consumer organization or by a third party, and may be hosted on the organization's premises (i.e. on-site private clouds) or outsourced to a hosting company (i.e. outsourced private clouds)



(a) On-site Private Cloud (b) Out-sourced Private Cloud

Benefits of choosing a Private Cloud

- The main benefit of choosing a private cloud is the greater level of security offered making it ideal for business users who need to store and/or process sensitive data.
- A good example is a company dealing with financial information such as bank or lender who is required by law to use secure internal storage to store consumer information.
- With a private cloud this can be achieved while still allowing the organization to benefit from cloud computing.
- Private cloud services also offer some other benefits for business users including more control over the server allowing it to be tailored to your own preferences and in house styles.
- While this can remove some of the scalability options, private cloud providers often

offerwhat is known as cloud bursting which is when non sensitive data is switched to a publiccloud to free up private cloud space in the event of a significant spike in demand until such times as the private cloud can be expanded.

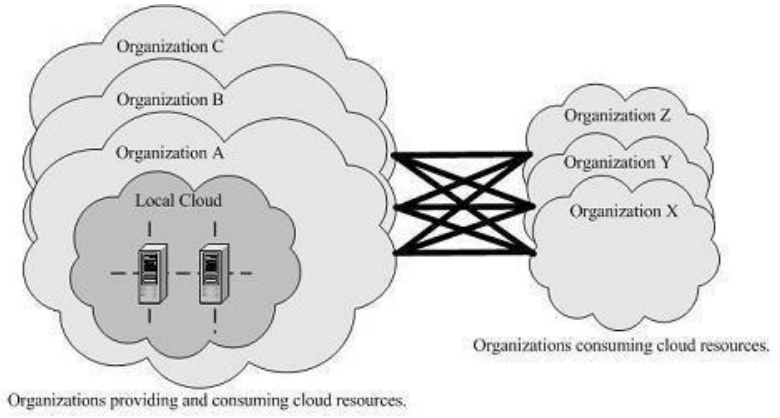
- In summary, the main benefits of the private cloud are:
    - Improved security
    - Greater control over the server
    - Flexibility in the form of Cloud Bursting

Disadvantages of choosing a Private Cloud

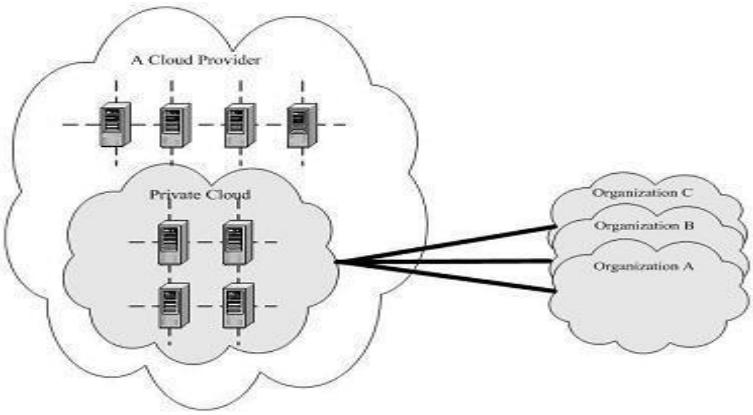
- The downsides of private cloud services include a higher initial outlay, although in the long term many business owners find that this balances out and actual becomes more cost effective than public cloud use.
- It is also more difficult to access the data held in a private cloud from remote locations due to the increased security measures.

Community Cloud

- A community cloud serves a group of Cloud Consumers which have shared concernssuch as mission objectives, security, privacy and compliance policy, rather than serving a single organization as does a private cloud.
- Similar to private clouds, a community cloud may be managed by the organizations or bya third party and may be implemented on customer premise (i.e. on-site community cloud) or outsourced to a hosting company (i.e. outsourced community cloud).
- A cloud consumer can access the local cloud resources, and also the resources of other participating organizations through the connections between the associated organizations.
- In this case, an outsourced community cloud builds its infrastructure off premise, and serves a set of organizations that request and consume cloud services.



(a) On-site Community Cloud



(b) Outsourced Community Cloud

Benefits of Choosing a Community Cloud

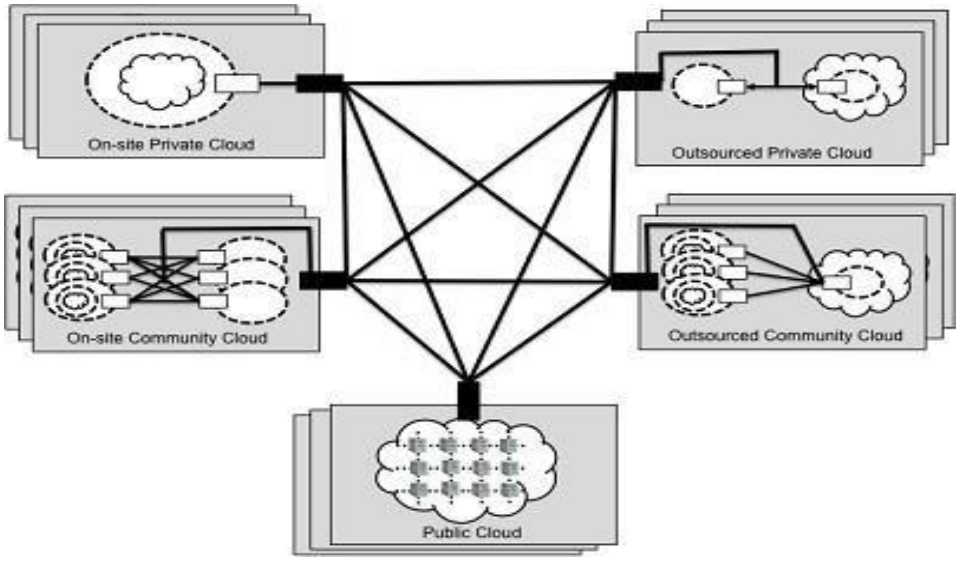
- Ability to easily share and collaborate
- Lower cost

Disadvantages of Choosing a Community Cloud

- Not the right choice for every organization
- Slow adoption to date

Hybrid Cloud

- A hybrid cloud is a composition of two or more clouds (on-site private, on-site community, off-site private, off-site community or public) that remain as distinct entities but are bound together by standardized or proprietary technology that enables data and application portability.
- Figure illustrates a simple view of a hybrid cloud that could be built with a set of clouds in the five deployment model variants.



Hybrid Cloud

Cloud Interoperability:

It is defined as the capacity of at least two systems or applications to trade with data and utilize it. On the other hand, cloud interoperability is the capacity or extent at which one cloud service is connected with the other by trading data as per strategy to get results.
The two crucial components in Cloud interoperability are usability and connectivity, which are further divided into multiple layers.

1. Behaviour
2. Policy
3. Semantic
4. Syntactic
5. Transport
6. Portability

It is the process of transferring the data or an application from one framework to others, making it stay executable or usable. Portability can be separated into two types: Cloud data portability and Cloud application portability.

- **Cloud data portability –**
  It is the capability of moving information from one cloud service to another and so on without expecting to re-enter the data.
- **Cloud application portability –**
  It is the capability of moving an application from one cloud service to another or between a client's environment and a cloud service.

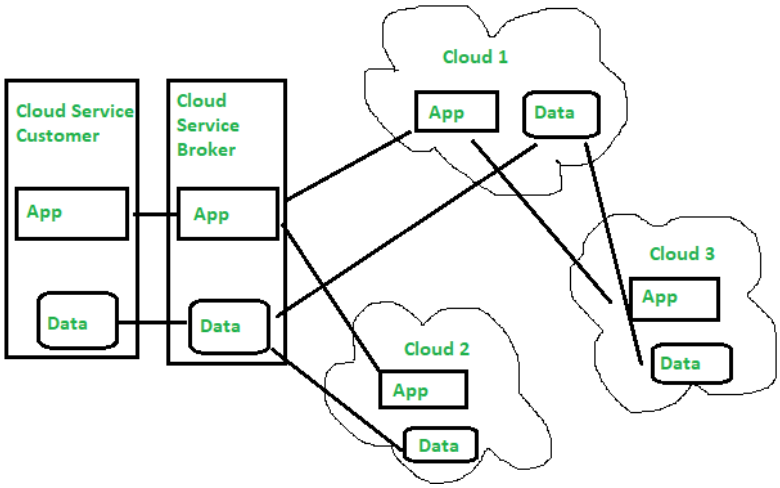
Categories of Cloud Computing Interoperability and portability:
The Cloud portability and interoperability can be divided into –

- Data Portability
- Platform Interoperability
- Application Portability
- Management Interoperability
- Platform Portability
- Application Interoperability
- Publication and Acquisition Interoperability

1. **Data Portability –**
   Data portability, which is also termed as cloud portability, refers to the transfer of data from one source to another source or from one service to another service, i.e. from one application to another application or it may be from one cloud service to another cloud service in the aim of providing a better service to the customer without affecting it's usability. Moreover, it makes the cloud migration process easier.
2. **Application Portability –**
   It enables re-use of various application components in different cloud PaaS services. If the components are independent in their cloud service provider, then application portability can be a difficult task for the enterprise. But if components are not platform specific, porting to another platform is easy and effortless.
3. **Platform Portability –**
   There are two types of platform portability- platform source portability and machine image portability. In the case of platform source portability, e.g. UNIX OS, which is mostly written in C language, can be implemented by re-compiling on various different hardware and re-writing sections that are hardware-dependent which are not coded in C. Machine image portability binds application with platform by porting the resulting bundle which requires standard program representation.

4. **Application Interoperability –**
   It is the interoperability between deployed components of an application deployed in a system. Generally, applications that are built on the basis of design principles show better interoperability than those which are not.
5. **Platform Interoperability –**
   It is the interoperability between deployed components of platforms deployed in a system. It is an important aspect, as application interoperability can't be achieved without platform interoperability.
6. **Management Interoperability –**
   Here, the Cloud services like SaaS, PaaS or IaaS and applications related to self-service are assessed. It would be pre-dominant as Cloud services are allowing enterprises to work-in-house and eradicate dependency from third parties.
7. **Publication and Acquisition Interoperability –**
   Generally, it is the interoperability between various platforms like PaaS services and the online marketplace.

The below figure represents an overview of Cloud interoperability and portability :



CLOUD INTEROPERABILITY AND PORTABILITY

Major Scenarios where interoperability and portability is required:
Cloud Standards Custom Council (CSCC) has identified some of the basic scenarios where portability and interoperability is required.

- **Switching between cloud service providers –**
  The customer wants to transfer data or applications from Cloud 1 to Cloud 2.
- **Using multiple cloud service providers-**
  The client may subscribe to the same or different services e.g. Cloud 1 and 2.
- **Directly linked cloud services-**
  The customer can use the service by linking to Cloud 1 and Cloud 3.
- **Hybrid Cloud configuration-**
  Here the customer connects with a legacy system not in a public, but, private cloud, i.e. Cloud 1, which is then connected to public cloud services i.e. Cloud 3.
- **Cloud Migration-**
  Clients migrate to one or more in-house applications to Cloud 1.

Challenges faced in Cloud Portability and Interoperability:

- If we move the application to another cloud, then, naturally, data is also moved. And for some businesses, data is very crucial. But unfortunately, most cloud service providers charge a small amount of money to get the data into the cloud.

Cloud Scalability

In cloud computing, cloud scalability refers to the ability to increase or reduce IT resources as required to meet evolving demands. Using existing cloud computing technology, data storage space, processing power and networking can all be escalated. Better still, scaling, usually with little or no interruption or downtime, can be achieved rapidly and easily. Third-party cloud providers now have the entire infrastructure in place; in the past, the process could take weeks or months to scale with on-site physical infrastructure and entail enormous costs.

To set up a personalized, scalable cloud solution via a public cloud, private cloud, or hybrid cloud, businesses have several options.

In cloud computing, two specific forms of scalability exist **vertical and horizontal scaling**.

We can add or subtract power to an existing cloud server memory upgrade, storage, or computing power with vertical scaling, also known as "scaling up" or "scaling down". This generally indicates that scaling has an upper limit based on the scaling capability of the server or machine; scaling above that also includes downtime.

We can add more resources like servers to our system using horizontal scalability to spread the workload across computers, which in turn improves efficiency and storage space. For companies with high-availability services that need limited downtime, horizontal scaling is essential.

Fault Tolerance in Cloud Computing

Fault tolerance in cloud computing means creating a blueprint for ongoing work whenever some parts are down or unavailable. It helps enterprises evaluate their infrastructure needs and requirements and provides services in case the respective device becomes unavailable for some reason.It does not mean that the alternative system can provide 100% of the entire service. Still, the concept is to keep the system usable and, most importantly, at a reasonable level in operational mode. It is important if enterprises continue growing in a continuous mode and increase their productivity levels.

- **Replication**: Fault-tolerant systems work on running multiple replicas for each service. Thus, if one part of the system goes wrong, other instances can be used to keep it running instead. For example, take a database cluster that has 3 servers with the same information on each. All the actions like data entry, update, and deletion are written on each. Redundant servers will remain idle until a fault tolerance system demands their availability.
- **Redundancy**: When a system part fails or goes downstate, it is important to have a backup type system. The server works with emergency databases that include many redundant services. For example, a website program with MS SQL as its database may fail midway due to some hardware fault. Then the redundancy concept has to take advantage of a new database when the original is in offline mode.

Objectives of Fault Tolerance in Cloud Computing

The fault-tolerant system uses backup components that take the place of failed components automatically, ensuring no service loss. They include:

Hardware systems
Hardware systems can be backed up using identical or equivalent systems. For instance, using an identical server running in parallel, with all operations mirrored to the backup server, a server can be made fault-tolerant.

Software systems
Software systems can be backed up using software instances. For example, it is possible to continuously replicate a database with customer information on another machine and operations can be mechanically redirected to another database in case a primary database goes down.

Power sources
Power sources use alternative sources using fault-tolerant. In many instances, organizations have power generators that can be used in case the electricity fails.
Similarly, using redundancy, any system or component that is a single point of failure can be made fault-tolerant.

Security Breach Occurrences
Owing to security failures, there are many explanations about why fault tolerance exists. The server's hacking adversely affects the server and results in a leak of data. Ransomware, phishing, virus attack, etc. are other explanations for the need for fault tolerance in the form of security violations.

Techniques for Fault Tolerance in Cloud Computing

- Priority should be given to all services while designing a fault tolerance system. Special preference should be given to the database as it powers many other entities.
- After setting the priorities, the Enterprise has to work on mock tests. For example, Enterprise has a forums website that enables users to log in and post comments. When authentication services fail due to a problem, users will not be able to log in.

Then, the forum becomes read-only and does not serve the purpose. But with fault-tolerant systems, healing will be ensured, and the user can search for information with minimal impact.

Major Attributes of Fault Tolerance in Cloud Computing

- **None Point of Failure**: The concepts of redundancy and replication define that fault tolerance can occur but with some minor effects. If there is no single point of failure, then the system is not fault-tolerant.
- **Accept the fault isolation concept**: the fault occurrence is handled separately from other systems. It helps to isolate the Enterprise from an existing system failure.

Existence of Fault Tolerance in Cloud Computing

- **System Failure**: This can either be a software or hardware issue. A software failure results in a system crash or hangs, which may be due to Stack Overflow or other reasons. Any improper maintenance of physical hardware machines will result in hardware system failure.

- **Incidents of Security Breach**: There are many reasons why fault tolerance may arise due to security failures. The hacking of the server hurts the server and results in a data breach. Other reasons for requiring fault tolerance in the **form of security breaches include ransomware, phishing, virus attacks, etc.**

Cloud Elasticity:

Elasticity refers to the ability of a cloud to automatically expand or compress the infrastructural resources on a sudden up and down in the requirement so that the workload can be managed efficiently. This elasticity helps to minimize infrastructural costs. It works such a way that when number of client access expands, applications are naturally provisioned the extra figuring, stockpiling and organization assets like central processor, Memory, Stockpiling or transfer speed.

It is most commonly used in pay-per-use, public cloud services. Where IT managers are willing to pay only for the duration to which they consumed the resources.

Example: Consider an online shopping site whose transaction workload increases during festive season like Christmas. So for this specific period of time, the resources need a spike up. In order to handle this kind of situation, we can go for a Cloud-Elasticity service rather than Cloud Scalability. As soon as the season goes out, the deployed resources can then be requested for withdrawal.

Difference Between Cloud Elasticity and Scalability:

	Cloud Elasticity	Cloud Scalability
1	Elasticity is used just to meet the sudden up and down in the workload for a small period of time.	Scalability is used to meet the static increase in the workload.
2	Elasticity is used to meet dynamic changes, where the resources need can increase or decrease.	Scalability is always used to address the increase in workload in an organization.
3	Elasticity is commonly used by small companies whose workload and demand increases only for a specific period of time.	Scalability is used by giant companies whose customer circle persistently grows in order to do the operations efficiently.
4	It is a short term planning and adopted just to deal with an unexpected increase in demand or seasonal demands.	Scalability is a long term planning and adopted just to deal with an expected increase in demand.

On-Demand Provisioning:

On-demand computing (ODC) is a delivery model in which computing resources are made available to the user as needed. The resources might be maintained within the user's enterprise or made available by a cloud service provider.

The on-demand business computing model was developed to overcome the challenge of enterprises meeting fluctuating demands efficiently. Because an enterprise's demand for computing resources can be unpredictable at times, maintaining sufficient resources to meet peak requirements can be costly. And cutting costs by only maintaining minimal resources means there are likely insufficient resources to meet peak loads. The on-demand model provides an enterprise with the ability to scale computing resources up or down whenever needed, with the click of a button.

Over-Provisioning:

Purchased resources are not fully utilized.

Under-Provisioning:

Purchased resources are not sufficient to meet the demands.

Resource Allocation Methods in Cloud Computing:

The allocation of resources and services from a cloud provider to a customer is known as resource provisioning in cloud computing, sometimes called cloud provisioning.

Resource provisioning is the process of choosing, deploying, and managing software (like load balancers and database server management systems) and hardware resources (including CPU, storage, and networks) to assure application performance.

To effectively utilize the resources without going against SLA and achieving the QoS requirements, Static Provisioning/Dynamic Provisioning and Static/Dynamic Allocation of resources must be established based on the application needs.

Resource over and under-provisioning must be prevented.

Importance of Cloud Provisioning:

- **Scalability:** Being able to actively scale up and down with flux in demand for resources is one of the major points of cloud computing
- **Speed:** Users can quickly spin up multiple machines as per their usage without the need for an IT Administrator
- **Savings:** Pay as you go model allows for enormous cost savings for users, it is facilitated by provisioning or removing resources according to the demand

Challenges of Cloud Provisioning:

- **Complex management:** Cloud providers have to use various different tools and techniques to actively monitor the usage of resources

- **Policy enforcement:** Organisations have to ensure that users are not able to access the resources they shouldn't.
- **Cost:** Due to automated provisioning costs may go very high if attention isn't paid to placing proper checks in place. Alerts about reaching the cost threshold are required.

Tools for Cloud Provisioning:

- Google Cloud Deployment Manager
- IBM Cloud Orchestrator
- AWS CloudFormation
- Microsoft Azure Resource Manager

Types of Cloud Provisioning:

- **Static Provisioning or Advance Provisioning:** Static provisioning can be used successfully for applications with known and typically constant demands or workloads. In this instance, the cloud provider allows the customer with a set number of resources. The client can thereafter utilize these resources as required. The client is in charge of making sure the resources aren't overutilized. This is an excellent choice for applications with stable and predictable needs or workloads. For instance, a customer might want to use a database server with a set quantity of CPU, RAM, and storage.
  When a consumer contracts with a service provider for services, the supplier makes the necessary preparations before the service can begin. Either a one-time cost or a monthly fee is applied to the client.
  Resources are pre-allocated to customers by cloud service providers. This means that before consuming resources, a cloud user must select how much capacity they need in a static sense. Static provisioning may result in issues with over or under-provisioning.
- **Dynamic provisioning or On-demand provisioning:** With dynamic provisioning, the provider adds resources as needed and subtracts them as they are no longer required. It follows a pay-per-use model, i.e. the clients are billed only for the exact resources they use. Consumers must pay for each use of the resources that the cloud service provider allots to them as needed and when necessary. The pay-as-you-go model is another name for this. "Dynamic provisioning" techniques allow VMs to be moved on-the-fly to new computing nodes within the cloud, in situations where demand by applications may change or vary. This is a suitable choice for programs with erratic and shifting demands or workloads. For instance, a customer might want to use a web server with a configurable quantity of CPU, memory, and storage. In this scenario, the client can utilize the resources as required and only pay for what is really used. The client is in charge of ensuring that the resources are not oversubscribed; otherwise, fees can skyrocket.
- **Self-service provisioning or user self-provisioning:** In user self-provisioning, sometimes referred to as cloud self-service, the customer uses a web form to acquire resources from the cloud provider, sets up a customer account, and pays with a credit card. Shortly after, resources are made accessible for consumer use.

Pros and Cons of Cloud Computing

Advantages

- Lower-Cost Computers for Users
- Improved Performance
- Lower IT Infrastructure Costs
- Fewer Maintenance Issues
- Lower Software Costs
- Instant Software Updates
- Increased Computing Power
- Unlimited Storage Capacity
- Increased Data Safety
- Improved Compatibility Between Operating Systems
- Improved Document Format Compatibility
- Easier Group Collaboration
- Universal Access to Documents
- Latest Version Availability
- Removes the Tether to Specific Devices

Disadvantages

- Requires a Constant Internet Connection
- Doesn’t Work Well with Low-Speed Connections
- Can Be Slow
- Features Might Be Limited
- Stored Data Might Not Be Secure
- Problem will arise If Data loss occurs